

Kesia Fernandes

510-770-4087 | kcafernandes06@gmail.com | linkedin.com/in/kesia-fernandes | github.com/kcafernandes

EDUCATION

University of California, Riverside

B.S. Robotics Engineering, Minor in Data Science

Sept. 2024 – June 2028 (Expected)

GPA: 3.8/4.0

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, HTML/CSS

Robotics & Embedded: ROS2, Arduino, Raspberry Pi, OpenCV, YOLOv8, SolidWorks

Libraries & Tools: pandas, NumPy, PyTest, GoogleTest, Valgrind

Dev & Workflow: Git, GitHub, Linux/Ubuntu, Agile/Scrum, Azure DevOps

EXPERIENCE

Software Developer — ACM Forge

Mar 2025 – Jun 2025

Association for Computing Machinery, UC Riverside

Riverside, CA

- Built and deployed a [Python](#)-based robotic arm system integrating [computer vision](#) and [serial communication](#) to control physical hardware in real time.
- Optimized the vision pipeline via [multithreading](#), cutting system latency by **30%** on resource-constrained [Raspberry Pi](#) hardware.
- Collaborated hands-on with mechanical and electrical subteams to integrate, validate, and iterate on autonomous control systems through multiple test cycles.

Technical Documentation Lead — VEX Robotics

Jun 2022 – Apr 2024

Competitive Robotics Team

Fremont, CA

- Authored engineering notebooks and [SolidWorks](#) CAD documentation covering hardware schematics, design iterations, and validation procedures across multiple build cycles.
- Coordinated software, mechanical, and manufacturing subteams to align design iteration, testing, and competition deliverables on tight timelines.

Robotics Instructor (Volunteer) — Friends of Children with Special Needs

Jun 2022 – Jun 2024

FCSN Youth Program

San Jose, CA

- Designed and delivered hands-on robotics curriculum using LEGO WeDo and Scratch for students with special needs.
- Created sensor-based and event-driven programming lessons, improving student engagement and foundational CS comprehension.

PROJECTS

Autonomous Robot Navigation | ROS2, Python, Linux/Ubuntu

Jan 2025 – Present

- Implemented [wall-following](#) and [trajectory generation](#) behaviors for a simulated robot using [ROS2](#) nodes, topics, and publishers/subscribers in a Linux environment.
- Continuing to explore [ROS2](#) simulation environments, building familiarity with autonomous navigation stacks and sensor integration.

Autonomous Robotic Arm | Python, YOLOv8, OpenCV, Raspberry Pi, Arduino

Mar 2025 – Jun 2025

- Designed, built, and deployed a vision-guided robotic arm using [YOLOv8](#) and [OpenCV](#) for real-time object detection and physical manipulation.
- Collaborated with mechanical and electrical engineers on [CAD](#) design and hands-on hardware integration, iterating through multiple test and validation cycles.

Text-Based Mystery RPG | C++, GoogleTest, Valgrind

Jun 2025 – Aug 2025

- Architected a C++ mystery game using [OOP](#) and [SOLID](#) principles; validated correctness and eliminated memory leaks with [GoogleTest](#) and [Valgrind](#).

AREAS OF INTEREST

Robotics & Automation, Embedded Systems, Autonomous Systems, Computer Vision, Sustainable Technology, Hardware-Software Integration